TEAM ASTRO

# AeroSentinel

## Satellite recovery decisions with protected mission history

9 task families / 3 difficulty levels / tool-using agents

A declared benchmark simulator

# Each action changes a persistent mission

| Layer | Agent sees or controls | Verifier evaluates |
|---|---|---|
| Observations | Manual, configs, mission targets and live telemetry | Seeded protected facts |
| Actions | Read, write, run Python, advance orbit, finish | Time, energy, exposure and delivered data |
| Limits | 40 replies and 10 tools per reply | 600 seconds plus a separate 120-second final check |

**The same tool and reward contract applies to every configured model.**

# Nine tasks, four planning challenges

| Group | Task families | Decision required |
|---|---|---|
| 5 guided repairs | Power, pointing, thermal, radio and fault cascade | Diagnose seeded faults |
| 2 radio missions | RF energy tradeoff<br>Early pass recovery | Balance link and energy<br>Meet a contact deadline |
| 2 payload missions | Observation scheduling<br>Science store and forward | Schedule useful exposure<br>Manage storage and contacts |

**Static nominal and maximum-radio policies fail all measured hard-task cases.**

# Two science windows, one finite buffer

| Mission stage | Required decision | Irreversible consequence |
|---|---|---|
| Before window 1 | Inspect targets and configure radio | Tool calls consume time |
| Window 1 → contact 1 | Collect, switch off, drain buffer | Lost data cannot be restored |
| Window 2 → contact 2 | Collect, switch off, deliver remainder | Both exposure quotas must pass |

**Constant duty needs ≥0.90 for exposure, but energy permits ≤0.101–0.152.**

Generated science = buffered + delivered + lost. Overflow must remain zero.

SCORING CONTRACT

# Reward follows verified mission history

| Raw score family | Standard | Observation | Science |
|---|---|---|---|
| Power / thermal | 0.35 / 0.25 | 0.28 / 0.20 | 0.21 / 0.15 |
| Pointing / comms | 0.20 / 0.20 | 0.16 / 0.16 | 0.12 / 0.12 |
| Exposure / science delivery | None | 0.20 / none | 0.20 / 0.20 |

**Progress = clamp((raw − no-op raw) / (1 − no-op raw), 0, 1)**

Every required check must pass. Fixed integration ticks remove timing artifacts.

# Separate processes protect scoring and mission facts

| Boundary | Authority | Protection |
| --- | --- | --- |
| Harness | Verifier, reward, mission truth | Separate container<br>Authenticated controls |
| Agent worker | Workspace repairs and tools | UID 1001, no outbound route<br>Read-only root and resource limits |
| Behavioral evaluator | Runs candidate calculation scripts | UID 1002, bounded fixtures<br>Worker cleanup between calls |

**Docker and the host kernel remain part of the trust boundary.**

# Six executed probes fail the success checks

| Executed probe | Reward | Outcome |
|---|---|---|
| Relaxed pointing target | 0.800 | failure |
| Forged telemetry | 0.000 | failure |
| Missed early contact | 0.418 | failure |
| Missed observation | 0.800 | failure |
| Missed science contact | 0.609 | failure |
| Forged science inventory | 0.521 | failure |

**Writable files cannot redefine protected quotas or manufacture collected data.**

# Reproducibility and shortcut comparison

**114 Linux tests** **180 task/seed pairs** **12 live isolation checks**

| Scripted policy | Successes / 45 | Evidence scope |
|---|---|---|
| No-op | 0 / 45 | 9 tasks × 5 separate seeds |
| Nominal / max radio | 25 / 45 and 20 / 45 | Both fail all 20 hard cases |
| Privileged reference | 45 / 45 | Feasibility baseline |

All 9 gates pass. Source and expected build inputs are checked before packaging.

# Four organizer-model validation attempts

| Attempt | Outcome / reward | Replies | Harness seconds |
|---|---|---|---|
| EPS Run A, no append | success / 1 | 11 | 49.44 |
| Radio energy, seed 101 | success / 1 | 11 | 58.97 |
| Science, seed 101 | success / 1 | 28 | 193.68 |
| Observation, seed 101 | failure / 0.8 | 12 | 94.39 |

**3 / 4 successful attempts. Full replies, actions and failure reasons are preserved.**

# Demo and presentation defense

## Run A: EPS recovery, seed 0

**success** **reward 1** **11 replies** **49.44 harness seconds**

## Defensible claims and limits

Coupled planning, deterministic resets, protected causal scoring.
Simplified physics, finite exploit coverage, one live deployment tested.

Public visibility conflicts with the PS. Access, name case and submission need confirmation.